

Increasing Array Queries

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array that consists of n integers. The array elements are indexed $1, 2, \dots, n$.

You can modify the array using the following operation: choose an array element and increase its value by one.

Your task is to process q queries of the form: when we consider a subarray from position a to position b , what is the minimum number of operations after which the subarray is increasing?

An array is increasing if each element is greater than or equal with the previous element.

Input

The first input line has two integers n and q : the size of the array and the number of queries.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Finally, there are q lines that describe the queries. Each line has two integers a and b : the starting and ending position of a subarray.

Output

For each query, print the minimum number of operations.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input	Output
5 3 2 10 4 2 5 3 5 2 2 1 4	2 0 14